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Enhancing Land Cover/Land Use (LCLU) classification through a comparative analysis of hyperparameters optimization approaches for deep neural network (DNN)

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ABSTRACT

Sustainable natural resources management relies on effective and timely assessment of conservation and land management practices. Using satellite imagery for Earth observation has become essential for monitoring land cover/land use (LCLU) changes and identifying critical areas for conserving biodiversity. Remote Sensing (RS) datasets are often quite large and require tremendous computing power to process. The emergence of cloud-based computing techniques presents a powerful avenue to overcome computing limitations by allowing machine-learning algorithms to process and analyze large RS datasets on the cloud. Our study aimed to classify LCLU for the Talassemtane National Park (TNP) using a Deep Neural Network (DNN) model incorporating five spectral indices to differentiate six land use classes using Sentinel-2 satellite imagery. Optimization of the DNN model was conducted using a comparative analysis of three optimization algorithms: Random Search, Hyperband, and Bayesian optimization. Results indicated that the spectral indices improved classification between classes with similar reflectance. The Hyperband method had the best performance, improving the classification accuracy by 12.5% and achieving an overall accuracy of 94.5% with a kappa coefficient of 93.4%. The dropout regularization method prevented overfitting and mitigated over-activation of hidden nodes. Our initial results show that machine learning (ML) applications can be effective tools for improving natural resources management.

1. Introduction

1.1. LCLU Monitoring for Environmental Management

Land Cover/Land Use (LCLU) classification can provide crucial insights into environmental dynamics on Earth (Faisal et al., 2021; Prasad et al., 2022a). As an active area of remote sensing (RS) research (Ekim and Sertel, 2021), LCLU monitoring enables change detection assessments, biodiversity analysis, and environmental surveying across large areas cost effectively compared to traditional methods. Specifically, land cover identification can be used to establish baseline conditions, while land use characterization can highlight socioeconomic conditions in built-up areas. Mounting anthropogenic pressures, such as rising global population and increasing land use transformations for agriculture, have exacerbated various environmental and climatic issues worldwide (Fei et al., 2018; Homer et al., 2020; Roy et al., 2022). Relating land cover spatially to land use can provide invaluable information to validate degradation processes, prioritize conservation efforts, and inform decision making at the local, regional, and national scale (da Jardim et al., 2021). Land use changes can also negatively impact biodiversity and ecosystem services for forest ecosystems (Azedou et al., 2022), which makes up a significant portion of the TNP (Ma et al., 2022; Modernel et al., 2016; Moukrim et al., 2018). Therefore, continuous LCLU monitoring is imperative to ensure the effectiveness of land use planning,

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especially in developing countries experiencing rapid development.

1.2. Emergence of ML and DL algorithms

Machine learning (ML) algorithms have gained popularity in recent years by improving classification accuracy and reducing processing costs relative to traditional methods (Yuh et al., 2023; Zerrouki et al., 2019). Prominent algorithms implemented for LCLU mapping include random forests (RF), support vector machines (SVM), artificial neural networks (ANN), k-nearest neighbors (KNN), decision trees (DT) and maximum likelihood (Al-Dousari et al., 2023; Chasmer et al., 2014; Choudhury et al., 2023; Eskandari and Ali Mahmoudi Sarab, 2022; Kafy et al., 2021; Kumari and Minz, 2023; Mollick et al., 2023; Prasad et al., 2022b, 2023; Rana and Venkata Suryanarayana, 2020; Shivakumar and Rajashekararadhya, 2018; Talukdar et al., 2021; Tan et al., 2021; Wu et al., 2021). Notably, the Sentinel-2 mission has fueled widespread LCLU applications utilizing these ML approaches. Previous studies report over 80% classification accuracy for crop/forest classes using RF algorithms on Sentinel-2 imagery (Inglada et al., 2015; Puletti et al., 2018). Conversely, SVM achieved 84.2% land cover mapping accuracy in Istanbul (Topaloğlu et al., 2016), while RF attained 65–76% accuracy for tree species/crop types (Immitzer et al., 2016). Sentinel-2 vegetation indices achieved 78% accuracy for conifer defoliation assessment via KNN, RF, and SVM (Hawryło et al., 2018). In particular, RF has stood out due to its ability to facility selecting interpretable features, ranking predictor importance, and expeditiously optimizing results (Amani et al., 2020; Azedou et al., 2021; Izquierdo-Verdiguier and Zurita-Milla, 2020; Wang et al., 2018a). Emergent techniques such as ANN and gradient-boosted algorithms also show promising results (Ebrahimy et al., 2022).

Deep learning (DL) techniques have also seen widespread applications across various RS use cases, including image preprocessing, scene classification, pixel-based classification, semantic segmentation, and target detection (Ma et al., 2019; Zhu et al., 2019). In particular, multilayer perceptron (MLP) as a fundamental DNN architecture has seen success in RS applications (Yuan et al., 2020; Al-Hameedi et al., 2021; Vinayak et al., 2021; Tripathi et al., 2022). DNN consisting of multiple fully connected layeremote can classify land cover from satellite imagery when provided suitable training data (Horry et al., 2023; Rumelhart et al., 1986). Recurrent neural networks (RNNs), adept at processing temporal data, have increasingly permeated RS literature for applications such as change analysis (Lyu et al., 2016). RNNs have also exhibited promising land cover classification performance where temporal transitions elucidate class identification, attaining 84.4% accuracy for crop mapping utilizing Sentinel-2 imagery (Rußwurm and Körner, 2017). RNN classification achieved even higher accuracy when used for a time-series analysis using Landsat-1 images (89%) (Sun et al., 2019) and for rice paddy delineation using Sentinel-1 images (98.3%) (Lin et al., 2022).

Integrating spatial and temporal DL dimensions remains an active area of research. A common approach has been to combine CNN and RNN models. Convolutional neural network (CNN)-RNN integration has manifested through multi-temporal architectures (Interdonato et al., 2019; Pelletier et al., 2019; Rußwurm and Körner, 2018) and parallel processing (Interdonato et al., 2019; Thorp and Drajat, 2021). Hybrid CNN-RNN fusion of Landsat, NAIP, climate and terrain datasets for land cover typing has shown promising results (Chang et al., 2019) while evaluating diverse spatiotemporal CNN-RNN designs assessing post-deforestation land use transition via Landsat further confirmed the combined models' superiority (Masolele et al., 2021). However, optimizing model hyperparameters remains imperative for leveraging these methods to address RS challenges. Model hyperparameters refer to adjustable parameters that control network architecture, optimization procedure, and regularization. Common hyperparameters include number of layers and nodes, learning rate, dropout rate, etc. However, optimizing model hyperparameters remains imperative for leveraging these techniques to address diverse RS challenges.

Reliable LCLU monitoring faces significant data-related challenges. Integration of multi-source datasets is often challenging due to mismatched temporal, spatial, and thematic resolutions (Ettehadi Osgouei et al., 2022). Temporally, land surveys often lack uniform periodicity, while historical records may have significant gaps (Nedd et al., 2021). Spatially, coarse-resolution satellite imagery cannot adequately resolve finer-scale land features that are important to landscape heterogeneity while cloud contamination can impede consistent records (Li et al., 2020). Thematic inconsistencies arise from subjective interpretation and shifting categorization schema, limiting comparability between datasets (Verburg et al., 2011), and the limited availability of ground-reference data for algorithm training and validation (Nedd et al., 2021).

Determining the optimal DNN architecture is an active area of research. Recent studies have evaluated different techniques for tuning hyperparameters, such as network structure, activation functions, regularization, and optimization algorithms (Xu and Liang, 2021). For instance, Lecun et al. (1998) demonstrated the effectiveness of stochastic gradient descent (SGD) for DNN training. Though it has been shown to get stuck in local minima, momentum was shown to help overcome this by accelerating SGD convergence (Luo et al., 2021). Dropout regularization has also proven essential for controlling overfitting as network depth and complexity increase (Gupta and Raza, 2020). More recently, exponential linear unit (ELU) activation functions have gained some traction for addressing ReLU limitations like dead neurons and slow convergence (Alkhouly et al., 2021; Clevert et al., 2016). Model optimization method comparisons have indicated that Bayesian optimization generally outperforms random search in efficiency and accuracy (Bergstra and Bengio, 2012; Snoek et al., 2012). Hybrid techniques that leverage complementary strengths have also emerged, such as combining Bayesian optimization and Hyperband methods (Wang et al., 2018b). Our study intends to contribute new insights towards optimizing DNN architectures to improve time-series LULC analyses.

1.3. Problem Statement and objectives of the Study

In northern Morocco, the Talassemtane National Park (TNP) faces significant challenges in managing its rich natural resources sustainably. The park supports high biodiversity and geodiversity, harboring many endemic plant species used in traditional medicine (Aoulad-Sidi-Mhend et al., 2019; Redouan et al., 2020), and geologically significant formations (Salhi et al., 2020). However, these are threatened by anthropogenic pressures (Benabid, 2008; Boukil, 1998; Jamila et al., 2018; Taheri et al., 2014). Interactions between coexisting tree taxa like Moroccan fir and Atlas cedar provide valuable insight into forest community dynamics (Ben-Said et al., 2022) but require accurate LCLU monitoring (Chemchaoui et al., 2023). Presently, available LCLU data for the park is limited, hampering effective conservation planning.

RS technologies offer the potential to generate up-to-date LCLU maps capable of detecting changes over time and identifying disturbance drivers (Avtar et al., 2017). With enhanced LCLU data, management authorities could assess impacts on vulnerable ecosystems and endangered species, monitor agricultural and infrastructure expansion, and develop targeted conservation policies founded on empirical evidence. Therefore, systematic monitoring of LCLU in the TNP using advanced RS approaches is needed to inform evidence-based conservation planning and resource management oriented towards the long-term preservation of its rich natural heritage. Addressing current data limitations strengthens adaptive ecosystem management in the face of ongoing threats. The objectives of this study were to (i) classify current (2022) LCLU in the park using a supervised Deep Neural Network (DNN) model, (ii) evaluate three hyperparameter optimization methods to identify the optimal DNN, and (iii) provide an updated LCLU map of the park. The updated LCLU map will help local authorities better understand land utilization in the park and implement sustainable environmental-related policies. Therefore, producing detailed and accurate LCLU data for TNP through optimized ML techniques is important for identifying priorities for conservation, monitoring LCLU changes over time, and informing policymaking to preserve the park's biodiversity for future generations.

1.4. Study area

The TNP was created in 2004, covering an area of 60,000 ha (El Gharbaoui, 1981), at the eastern end of the limestone ridge of the Rif Mountains (geographical coordinates 35°13′31″ N/5°08′26″ W) (Fig. 1). The park houses biological, ecological, and landscape diversity with important heritage value (Redouan et al., 2020). This site is characterized by hot, dry summers and cool winters, with average annual temperatures ranging from 14 to 20 °C and average annual precipitation often exceeding 1000 mm (ABHL, 2006).

The calcaro-dolomitic massif of Talassemtane is a particular hotspot of biodiversity. It includes many fir tree species that are endemic to North Africa (Abies maroccana, Cedrus atlantica, *Pinus nigra* subsp. mauritanica, Acer granatensis, Quercus rotundifolia, *Pinus pinaster* subsp. Maghrebiana) (Aafi et al., 1997) along with 1380 plant species, 33 species of amphibians and reptiles, 180 species of aquatic macroinvertebrates, and about 40 species of mammals (Rapport MEDA, 2008).

Agricultural activities, such as arboriculture, beekeeping, and tourism, are the main socio-economic activities in the region. The forest area within the park is threatened by clearing and plowing activities for new cannabis cultivation and illegal logging and hunting violations. Moreover, the park has witnessed a significant increase in the number of fires, most of them intentional, that have destroyed an average of 500 ha per year (Bosshardt, 2007).

2. Methods

2.1. Dataset preparation and environment of work

Our study examines a DNN model for supervised LCLU classification in a cloud-based environment. The multispectral image dataset was obtained from Sentinel-2 MSI Level 2-A (10 m resolution) for 2022, and the vector dataset used, including ground truth data, was provided by local managers and validated by visits to the NPT territory. The LCLU classification was conducted using Python programming language (version 3.7.13).

2.2. Land Cover/Land Use (LCLU) classification

LCLU classification was performed using Google Earth Engine (GEE) as the geospatial processing platform and satellite imagery catalog, Google Colab as the online framework to build, execute, and share our ML and DL models, and Google Cloud to store the data and model for improved accessibility (Fig. 2).

2.3. Image preprocessing for region of interest (ROI) filtering

A cloud-masking algorithm (cloud cover less than 10%) was applied to the Sentinel-2 image collection to mask obstructions from cloud and shadow pixels to retain unobstructed pixels. A median reducer was then applied to aggregate the remaining pixels and the image collection was

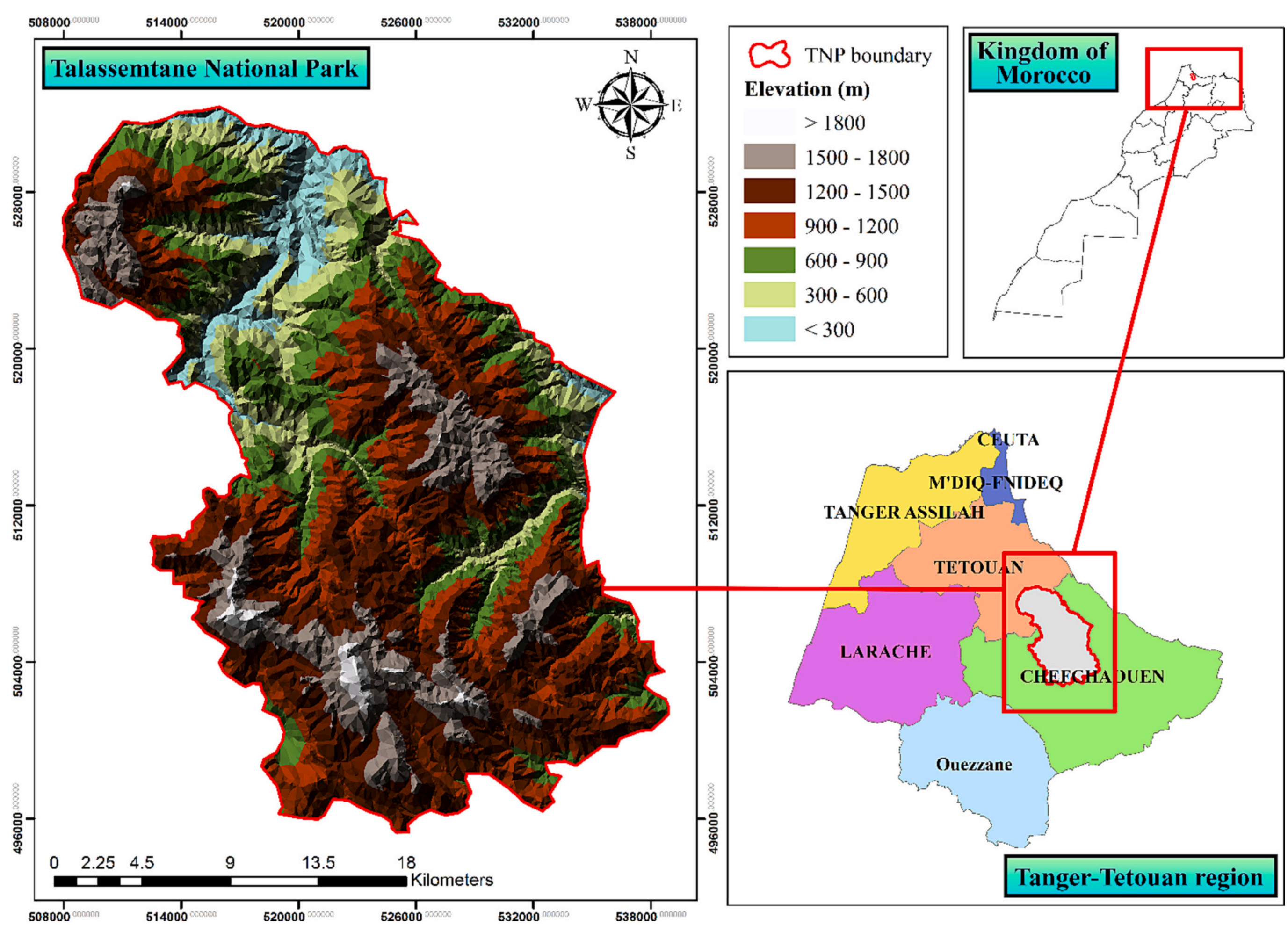

Fig. 1. Location of the National Park of Talassemtane.

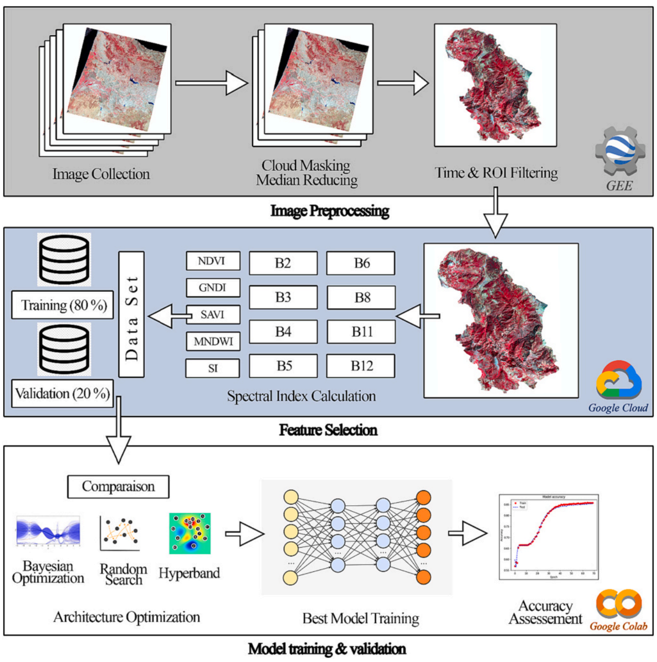

Fig. 2. Technical flow chart of the LCLU classification.

then processed to generate a single composite image for June–September 2022. The final composite image was subdivided into 256 × 256-pixel patches and exported to the cloud. The training data was parsed to reduce variability (as a float32 format) for efficient processing.

2.3.1. Reference data set preparation

A sample reference database for the main LCLU classes was prepared using visual interpretation of high-resolution Google Earth imagery. To validate and correct the interpreted LCLU classes, approximately 1000 of the 3000 reference points were surveyed in the field by five members of the research team. The remaining points were verified through RS interpretation with the help of National Agency of Water and Forests agents. The validated points were then compared to the initial classes assigned through RS interpretation, and corrections were made as needed.

The 3000 reference points represented six LCLU classes: forest, matorral, agricultural lands, built-up areas, limestone soil surfaces, and bare lands (Fig. 3).

The points were randomized and divided into training (80%) and validation (20%) sets. A 15 m buffer was applied around all sample points to ensure independence between the datasets (Fig. 4). The reference points were then exported to the cloud for model training and validation.

2.4. Indexes calculation

The Sentinel-2 surface reflectance bands (B2: Blue, B3: Green, B4: Red, B5: Red Edge 1, B6: Red Edge 2, B7: Red Edge 3, B8: NIR, B11: SWIR 1, B12: SWIR 2) were used to construct the initial feature set. Five RS indices were added to the training data set to guarantee an accurate classification. RS offers a wide range of indices that can be used for LCLU classification, such as the normalized difference vegetation index (NDVI). NDVI has been used for change detection in several fields such, as forest disturbances and recovery (Neigh et al., 2014; Wang, 2012), crop monitoring (Seo et al., 2019), environmental change monitoring (Jiang et al., 2021) and, climate change (Baniya et al., 2018).

The NDVI index is computed as:

Fig. 3. Representative landscape photos of each LCLU class in the study area.

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

where *NIR* and *Red* are the reflectance values from the Sentinel-2 bands B8 (835.1–833 nm) and B4 (664.5–665 nm), respectively.

Given the intense agricultural activity in the PNTLS, accurate cropland identification is essential. The green normalized difference vegetation index (GNDVI) is a reliable indicator of crop canopy vigor (Khot et al., 2016; Zhou et al., 2018) and is characterized by a linear correlation between leaf area index (LAI) and biomass, making it more sensitive to biomass and chlorophyll concentration changes than NDVI (Gitelson et al., 1996). The GNDVI index is calculated as:

$$GNDVI = \frac{NIR - Green}{NIR + Green}$$

where *NIR* and *Green* are reflectance values in the near infrared and green bands corresponding to the Sentinel-2 bands B8 (835.1–833 nm) and B3 (560–559 nm), respectively.

NDVI can be influenced by soil reflectance in areas with low vegetation cover and exposed soil. Therefore, we included a soil-adjusted vegetation index (SAVI) to correct soil influence using a soil brightness correction factor (L). The SAVI index is calculated as (Huete, 1988):

$$SAVI = \frac{(NIR - Red)}{(NIR + Red + L)} * (1 + L)$$

where *NIR* and *Red* are reflectance values of the Sentinel-2 bands B8 (835.1–833 nm) and B4 (664.5–665 nm), respectively. L is the adjustment factor that varies with vegetation density, ranging from 1 for very low densities to 0.25 for denser canopies (Gonzalez-Piqueras et al., 2004). A value of 0.5 was set as the vegetation density ranges from dense to sparse.

The modified normalized difference water index (MNDWI) was also added to improve the identification of water bodies, especially retention ponds used to grow annual crops in the PNTLS. Developed by Xu (2006), the MNDWI can distinguish water bodies without being sensitive to built-up areas. The MNDWI is calculated as:

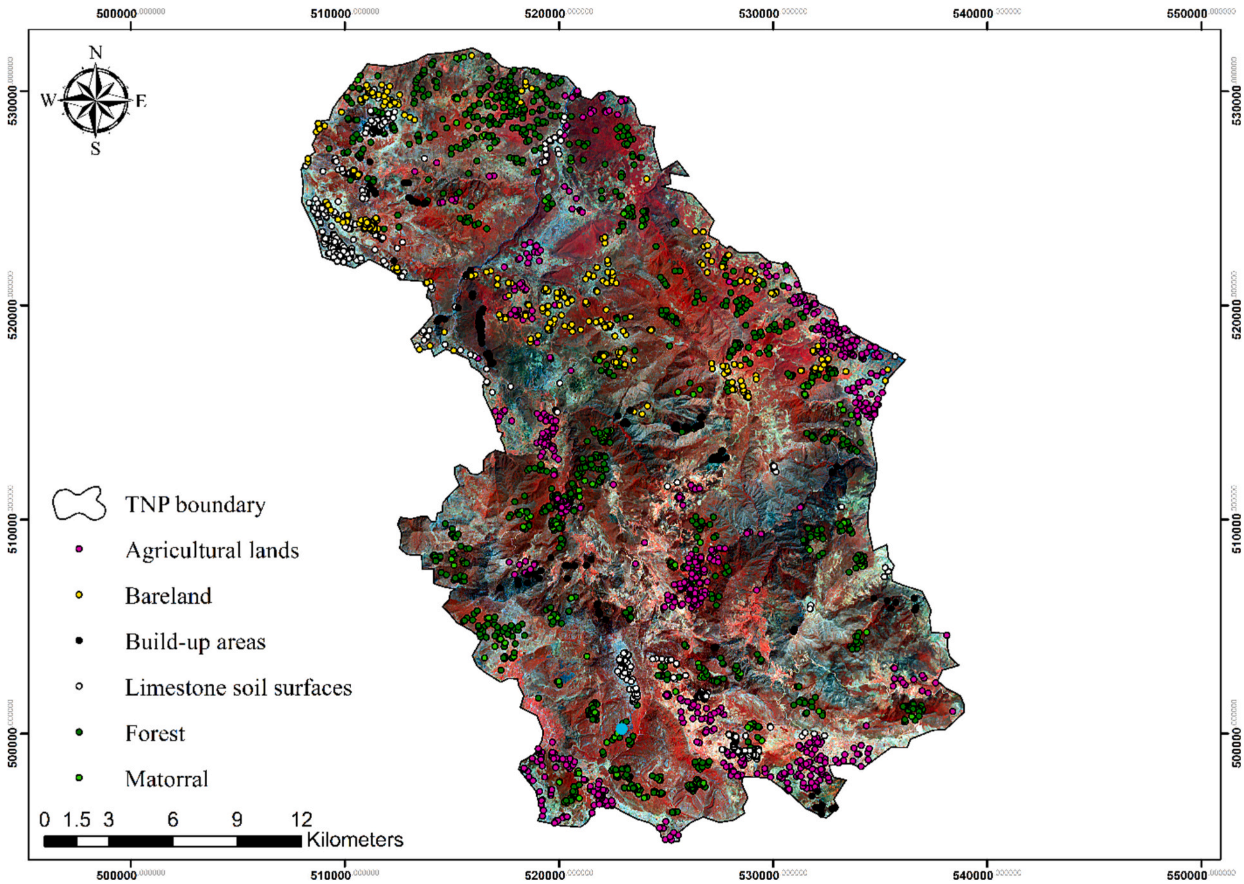

Fig. 4. Spatial distribution of field reference sample points collected across the study area.

$$MNDWI = \frac{Green - SWIR}{Green + SWIR}$$

where *SWIR* and *Green* are the reflectance values of the Sentinel-2 bands B11 (560–559 nm) and B3 (1613.7–1610.4 nm), respectively.

Lastly, a shadow index (SI) was added to detect dense stands such as cedar and pine forests highlands. This helped identify shadow patterns that affected the spectral response of the forest canopy. Moreover, SI played an essential role in determining the boundary of vegetation change between different forest formations and vegetation biomass estimation (Ono et al., 2010). SI is computed as (Rikimaru et al., 2002):

$SI = (\sqrt[3]{(256 - Red)^*(256 - Green)^*(256 - Blue)})$.

where *Red, Green* and, *Blue* are the reflectance values of the Sentinel-2 bands B4 (664.5–665 nm), B3 (560–559 nm) and, B2(496.6–492.1 nm), respectively.

2.5. Classification algorithm

2.5.1. Deep Neural Network (DNN)

Regular DNN or MLP is based on the concept of the original perceptron model proposed by Rosenblatt (1958) and features an architecture comprising four main components: (1) an input layer, (2) neurons (nodes), (3) hidden layers, and (4) an output layer. DNN can be described as a layout of neurons grouped into layers and linked to each other by weights, which are iteratively adjusted by an optimization process (Jozdani et al., 2019). The data is initially presented to the input layer, which distributes them to each of the neurons in the first hidden layer. Outputs from the first hidden layer are then distributed to the following hidden layers until they reaches the output layer, where the data is distributed to output neurons equaling the number of classes.

2.5.2. Backpropagation learning

DNN uses the backpropagation algorithm for the training process. The network output is compared to the expected output to calculate the error. The latter will be propagated back to update the weights using a stochastic gradient descent approach for loss function optimization. This process was repeated iteratively, with a round for updating the entire network for all the training data, called an epoch.

For a set of hidden layers $(h_1, h_2, \dots, h_n)$ where n_i represents the neuron number for each hidden layer h_i, the output for the first hidden layer is calculated as:

$$h_i^j = f\left(\sum\nolimits_{k=1}^{n_{i-1}} w_{k,j}^0 x_k\right); j = 1, \dots, n_i$$

The output-hidden layer h_i^j is calculated as:

$$h_i^j = f\left(\sum\nolimits_{k=1}^{n_{i-1}} W_{k,j}^{i-1} h_{i-1}^k\right) \; i = 2, \dots, N \; and \; j = 1, \dots, n_i$$

where $W_{k,j}^i$, is the weight between the neuron k in the hidden layer i and the neuron j in the hidden layer $+1$, n_i is the number of the neurons in the hidden layer i.

The output of the hidden layer i is computed as:

$$h_i = {}^t\left(h_i^1, h_i^2, \dots, h_i^{n_i}\right)$$

The output layer is defined as:

$$y_i = f\left(\sum_{k=1}^{n_N} w_{k,j}^N h_N^k\right); Y = (y_1, \dots, y_j, \dots, y_{N+1}) = F(W, X)$$

where $w_{k,j}^N$ is the weight between the neuron k in the N^{th} hidden layer and the neuron j in the output layer, n_N is the number of the neurons in the N^{th} hidden layer, Y is the output layer vector, F is the transfer function and w is the matrix of weights calculated as:

$$W = \left[W^0, \dots, W^j, \dots, W^N\right]; W^i = \left(w^i_{j,k}\right) \begin{matrix} 0 \le i \le N \\ 1 \le j \le n_{i+1} \\ 1 \le k \le n_i \end{matrix} \; where \; w^i_{j,k} \in \mathbb{R}$$

The input data is processed by an activation function, which is generally non-linear. The latter introduces non-linearity into the network, enabling it to learn from complex patterns and model relationships between input and output data. This allows for a complex combination of input data and provides robust modeling capability (Botero-Valencia et al., 2019; Mellit and Kalogirou, 2008).

2.6. DNN architecture optimization strategy

The model architecture is defined by the number of layers and neurons used for each layer. The arrangement of these two parameters strongly affects the predictive qualities of the model and it is a critical question that needs to be addressed in classification use cases (Ghanou and Bencheikh, 2016; Gupta and Raza, 2019). Currently, no specific rules govern the optimal structure for a neural network (Zhang et al., 2019). However, single-layer neural networks are used exclusively to represent linearly separable functions, while multiple layers allow for greater depth that results in better generalization for more complex problems (Goodfellow et al., 2016).

The best architecture is determined using a “trial and error” approach, consisting of testing and evaluating several combinations of hyperparameters, where the most accurate combination is selected for the final model. The literature shows various methods for optimal estimation of neural network structure (Stathakis, 2009); however, from heuristic searches to exhaustive strategies, trial and error methods remain the most common (Ali et al., 2017).

This study adopted an automated optimization architecture by comparing three architecture optimization algorithms: Random Search (RS), Hyperband, and Bayesian optimization. Hyperparameters of our DNN model were automatically optimized according to a search grid of several hyperparameters listed in Table 1. A dropout layer was added to prevent overfitting and to reduce statistical noise. The dropout regularization randomly removed different sets of neurons using a predetermined dropout rate, which resulted in training different neural networks and reduced overfitting (Srivastava et al., 2014).

2.6.1. Random search

Random Search (RS) is an optimization method widely used in ML and artificial intelligence (AI) applications. It is a simple and effective optimization approach that is often used as a baseline for comparing other optimization methods. A set of hyperparameters is selected and evaluated randomly to find the set with the best accuracy for a random search optimization. This method suits ill-structured global optimization problems where the objective function may be non-convex and non-differentiable (Zabinsky, 2009). RS can outperform grid search and compete with Bayesian optimization for a classification problem on multiple MNIST handwritten digit data sets (Bergstra and Bengio, 2012). In addition, RS can solve large-scale problems in a way that is not possible with deterministic algorithms (Vavasis, 1995).

Table 1
Configuration of tested hyperparameters.

Hyperparameters	Configuration		
	Min value	Max value	Step
N° of neurons	2	512	2
N° of hidden layers	2	128	2
Dropout	0.0	0.5	0.05
Activation functions	Rectified Linear Unit, Exponential Linear Unit, Scaled Exponential Linear Unit, Sigmoid, Softmax, Softplus, Softsign, Hyperbolic Tangent, Exponential, Scaled		
Optimizer	Stochastic Gradient Descent (SGD)		

2.6.2. Hyperband

The Hyperband optimization method (Li et al., 2018) became a popular optimization method for building ML models because it reduced computational costs by using several rounds of a bandit-based algorithm called Successive Halving (Jamieson and Talwalkar, 2015) to explore the hyperparameter search grid efficiently. The best hyperparameter configuration is then selected after eliminating the worst-performing configurations.

The main advantage of Hyberband is that it discards poorly performing configurations early in the learning process to reduce the computational cost of model training and evaluation while improving classification accuracy. This is done using the early stopping technique, which allows the algorithm to evaluate the performance of a configuration after only a small fraction of the total training time has elapsed since it has been shown that many ML models can still achieve good performance early in the training process. In some cases, it has even offered better results than state-of-the-art algorithms, such as Bayesian optimization and grid search (Falkner et al., 2018). The Hyperband method also improved classification accuracy significantly compared to other optimization methods for image recognition and reinforcement learning tasks. (Jamieson and Talwalkar, 2015; Li et al., 2018).

2.6.3. Bayesian optimization

Unlike the grid and RS methods, which entirely ignore previous evaluations, the Bayesian optimization algorithm has the advantage of keeping track of previous results to choose the best hyperparameters. For a model Z and an observation Y, the optimization procedure is calculated using the following equation (Kramer et al., 2011):

$$P(Z|Y) = \frac{P(Y|Z)^{*}P(Z)}{P(Z)}$$

where $P(Z|Y)$ is the posterior probability of Z given Y, $P(Z|Y)$ is the likelihood of Y given Z, $P(Z)$ is the prior probability of Z, and $P(Z)$ is the marginal probability of Z. This method relies on Baye’s Theorem (Kramer et al., 2011) to overcome the necessity of a large number of learning cycles. It shows excellent potential for handling noisy data, adapting to maximum resource utilization, and exploiting non-continuous spaces to achieve global minima (Victoria and Maragatham, 2021). Furthermore, it has the advantage of selecting the next set of hyperparameters in a guided way, making it more efficient than other grid or RS methods (Yu and Zhu, 2020).

2.7. Accuracy assessment

The model with the best accuracy was retained after training the three main DNN models for 200 epochs. An overall accuracy measure was conducted to evaluate the accuracy level of our final DNN model using our validation dataset. A confusion matrix was produced to highlight the accuracy of each class. Three quality metrics were added: Overall Accuracy (OA), Producer Accuracy (PA), and User Accuracy (UA).

Overall accuracy is the ratio of the sum of correctly classified pixels to the total number of pixels in the test dataset. While the producer accuracy refers to the accuracy from the point of view of the mapmaker, the user’s accuracy represents the probability that a pixel classified in each category corresponds to the actual category on the ground. The performance of our final DNN model was evaluated using the kappa coefficient of agreement (Landis and Koch, 1977):

$$I = \frac{n\sum_{i=1}^{c} x_{ii} - \sum_{i=1}^{c} x_{i+}x_{+i}}{n^2 - \sum_{i-1}^{c} x_{i+}x_{+i}}$$

where, I is kappa coefficient; n is total number of observations; c is the total number of classes; x_{ii} is the value in row i and column i, x_{i+} is the sum of row i and x_{i+} is the sum of the column i of the matrix.

3. Experimental results

3.1. DNN Architecture optimization

The optimization process considers three main factors. The main factor is the model's accuracy when tested using the validation dataset. Computation time and costs is the second deciding factor, as both can add up over time with the increasing complexity of models and datasets. Efficient methods for hyperparameter optimization are necessary to obtain high-performing models while leading to more efficient utilization of computational resources. Lastly, commonalities and differences between hyperparameters can better highlight the impact of each hyperparameter on the model's performance.

A search space of the primary hyperparameters was defined. Optimization algorithms were executed to find the highest accuracy from 1,788,160 hyperparameter combinations. Our model was trained for 200 epochs for each trial to find three DNN architectures that were used to build three models and train them for the entire dataset for 1000 epochs (Table 2).

Results from our comparison showed that each method produced its unique architecture for the MLP model. RS provided the largest architecture with 34 hidden layers; however, randomly identifying the best hyperparameter combination required additional time. While this method removed the dropout layer, Hyperband and Bayesian optimization methods included a dropout layer with dropout rates of 0.05 and 0.15, respectively. The two main differences between the models were the number of hidden layers and the activation function. The Hyperband method proved to be the most efficient, having the lowest computation time compared to the other methods. A schematic representation of the architecture for each model is depicted in Fig. 5.

Overall accuracy and computation time results for the Random Search, Hyperband, and Bayesian optimization methods are illustrated in Fig. 6.

The findings show no trade-offs between the optimization algorithms' run time and overall accuracy. Hyperband was the fastest algorithm, with a run time of 5 h and 27 min, and achieved the highest overall accuracy of 94.5%. On the other hand, Bayesian optimization achieved an accuracy of 89.4%, slightly lower than Hyperband but still significantly higher than RS's accuracy of 82.8%. Bayesian optimization also took longer than Hyperband but was still faster than RS. The Hyperband optimization method provided better accuracy than the RS

Table 2
Statistics of DNN optimized architecture using Random Search, Hyperband and Bayesian Optimization methods.

Architecture	Random Search	Hyperband	Bayesian Optimization
Number of hidden layers	34	32	8
Min number of neurons	8	6	6
Max number of neurons	118	128	118
Average number of neurons	56	75	20
Dropout	0	0.05	0.15
Activation Function	Exponential Linear Unit	Exponential Linear Unit	Rectified Linear Unit

or Bayesian optimization approaches and should, therefore, be used if sufficient computation resources are available.

Learning and loss curves were also used to diagnose the learning performance of the three models over time (Fig. 7). The training accuracy curve provides insight into how the model is learning, while the validation accuracy gives insight into the generalizing process. Likewise, the training loss curve shows the model's performance on the training data when the optimization algorithm adjusts the model parameters to minimize the loss function. The validation loss curve measures the model's performance on new unseen data.

Overall, the accuracy and loss curves of the Hyperband and Bayesian optimization methods show similar patterns. Both methods had a steep increase in accuracy and a steep decrease in loss function. This indicated that both models learned quickly and fitted the training and validation data well. Furthermore, both models' accuracy and loss values plateaued after a certain point, indicating that they converged to a good solution.

The Hyperband method plots showed a non-smooth pattern with irregular values. This "unstable" learning reveals that the model had difficulty learning the data patterns. In the loss plot, the training loss curve and validation loss curve crossed, which means that the model performed differently on the training dataset than on the validation dataset. By excluding a regulation technique, this model overfitted by showing good performance in the training dataset but failing to generalize onto the validation dataset. Overfitting occurs when the model is too complex and captures noise in the training data instead of the underlying patterns. Even though the Hyperband and Bayesian optimization methods showed similar learning behaviors, they did not perform similarly. The training and validation accuracy curves are closer to the Hyperband than the Bayesian optimization method. The two curves show a difference of 0.03% and 0.06% for the Hyperband and the Bayesian optimization method, respectively. However, for the loss curves, the opposite was true. The two-loss curves showed a difference of 0.1% and 0.05 for the Hyperband and the Bayesian optimization method, respectively. The difference between the accuracy of the validation and training curves is linked to the model's poor performance. Since the Hyperband showed the slightest accuracy difference between the training and validation curves, we considered it the best architecture optimization approach.

3.2. Classification results

The box plot shown in Fig. 8 shows the spectral reflectance and computed index values of 500 randomly selected samples generated for each LCLU class type from the classified image. Note that the SI values were normalized to facilitate comparison with other vegetation indices. The box plots figures of band reflectance showed either no or significant differences in reflectance among each class. Those with significant differences can be divided into either vegetation classes or other classes (e.g., human construction, bare lands, limestone surfaces, etc.)

Bands 2 and 5 showed that the agricultural lands class had a significant spectral difference compared to forest and matorral classes for vegetation classes. On the other hand, Bands 11 and 12 provided better separation between the forest and matorral classes. While Bands 3, 4, 5, and 11 had clear spectrum differences between vegetation classes and other classes, distinguishing spectral values for those two groups was difficult in Bands 6 and 8. Finally, the reflectance of the limestone surfaces and build-up areas had the same pattern, while bare surfaces displayed a slightly lower reflectance.

Including spectral indices, such as NDVI, MNDWI, GNDVI, SAVI, and SI, can improve classification results between LCLU classes with similar reflectance. NDVI index showed good separability between vegetation classes and other classes; however, the spectral difference between the MNDWI values for the vegetation classes was very minimal. Using the GNDVI and SAVI indices can overcome this issue by better detecting forest, matorral and agricultural lands. However, the latter showed low

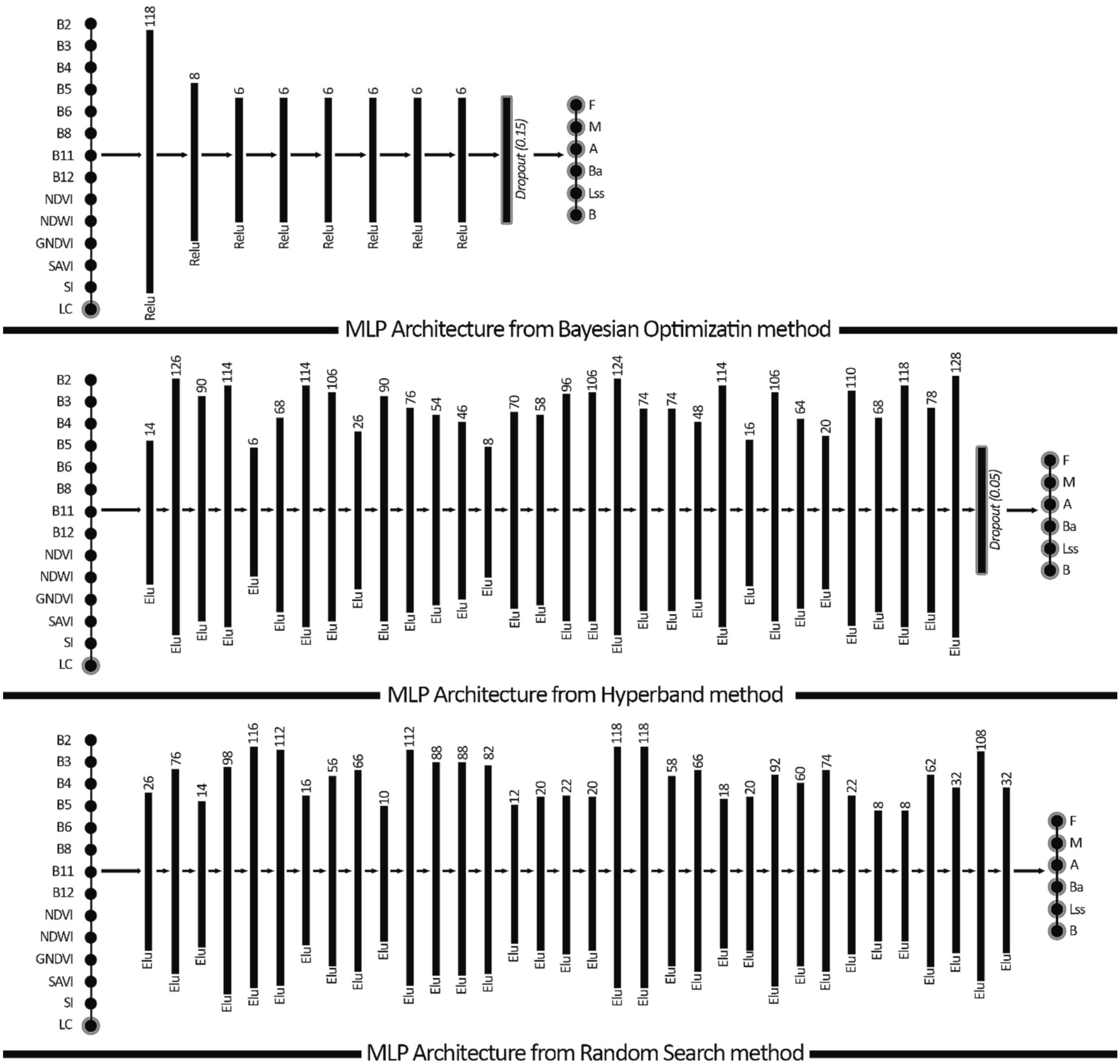

Fig. 5. DNN architecture from Random Search, Hyperband and Bayesian Optimization methods.

reflectance compared to other LCLU classes for most indices except SI, which led to low identification of agricultural areas. Built-up areas showed very high MNDWI values compared to other classes, allowing for better classification as it still remained difficult to identify, especially in our study where the human construction in villages covered small areas. The SI values also showed a significant difference between bare land, build-areas and, other classes due to the absence of trees.

All thirteen training features (eight Sentinel-2 spectral bands and five indices) were used in the classification process to generate the 2022 TNP land use map (Fig. 9). We highlight three typical areas for a detailed illustration of the results (Fig. 10).

The land cover classification allowed us to identify the central forest massifs within the park. Forest areas were dominated by holm oak, cork oak, atlas cedar, maple, beech, ash, and juniper. Matorral areas comprised various shrubs, herbs, and bushes, such as thyme, rosemary, laurel, broom, heather, and rockrose. The farming systems were mainly used for traditional crops such as grains, fruits, and vegetables. Farmers in the area also grew cannabis for local consumption and export. Built-up areas were presented in small settlements and grouped in clusters or neighborhoods. Total area (ha) for each LCLU class is presented in Table 3.

The data acquired in the field visits facilitated recognizing our study area's primary land use activities. Distinguishing between similar classes was challenging, especially for built-up areas that stood out as small settlements. The classification of agricultural lands showed improved results, even though most of the agricultural land remains uncultivated before the planting period.

3.3. DNN Accuracy assessment

The test sample was used to evaluate the accuracy of the classification results using the confusion matrix (Fig. 11). The model showed

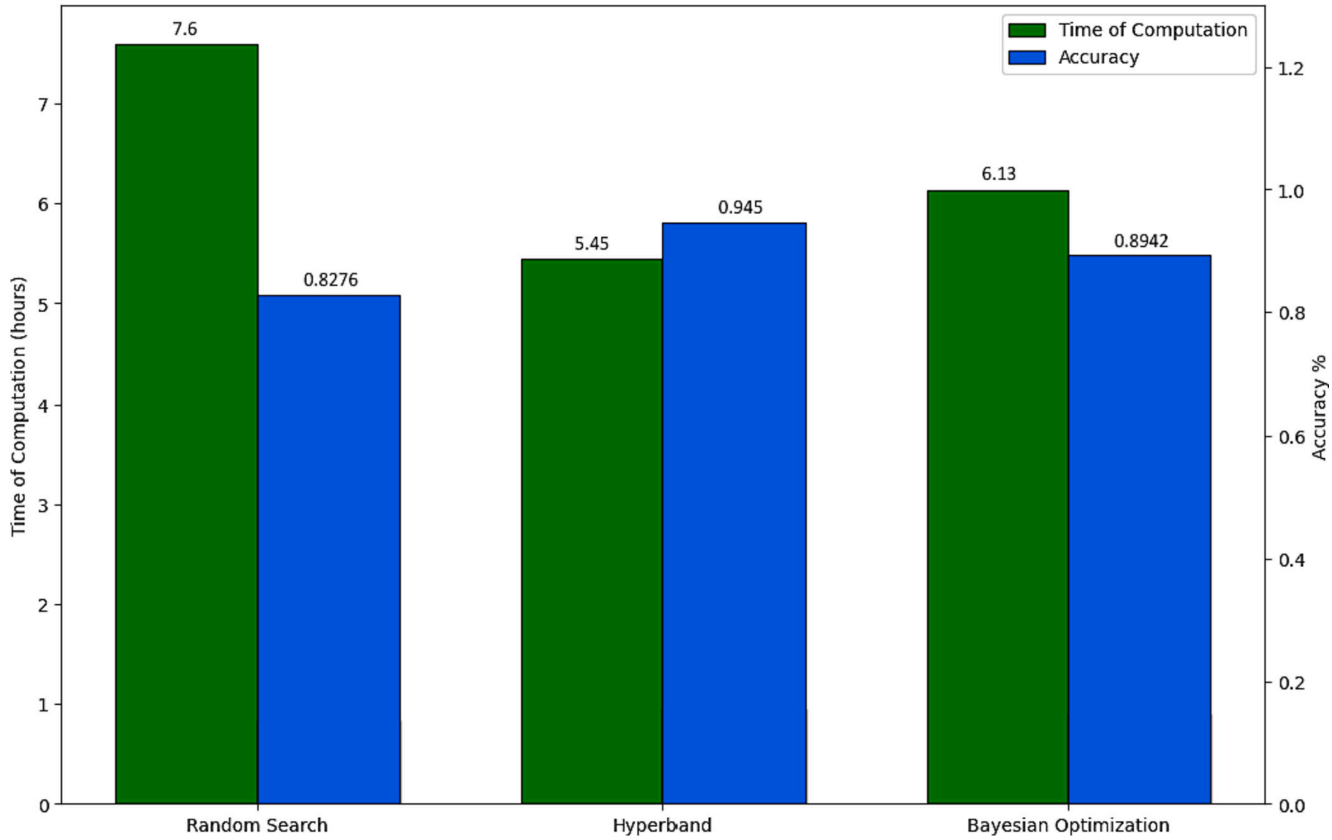

Fig. 6. Overall accuracy and time of computation for our optimization methods.

excellent predictive capacity with an overall accuracy of 94.5% and an almost perfect agreement with a kappa coefficient of 93.4%.

To better assess the performance of our DNN model for each class, errors of omission and commission were calculated for each LCLU class (Fig. 12). Errors of omission represent the proportion of reference data pixels that the model incorrectly classifies. In contrast, errors of commission correspond to the proportion of model-predicted pixels that were incorrectly classified according to the reference data. These accuracy metrics should be considered together when evaluating the model's performance. A model with high errors of omission may miss important observations, while a model with high errors of commission may produce many false positives. The balance between these two types of errors depends on the specific application and the cost of each type of error to each respective study.

Our results show that the error of omission was lowest for the agricultural lands (4.16%), indicating that the model performed relatively well when predicting this class. The error of commission was highest for the built-up areas class (8.42%), signaling that the model was more likely to incorrectly predict an observation belonging to the built-up areas class. The bare land class had the same error of omission and error of commission (3.09%), indicating that the model performs equally well in correctly predicting avoiding false positives for this class.

4. Discussion

4.1. LCLU classification from Sentinel-2 satellite imagery

This study aimed to perform a detailed LCLU classification of Morocco's Talassemtane National Park using advanced ML techniques to process satellite imagery. A DNN model was developed and trained on Sentinel-2 multi-spectral images to classify six primary LCLU classes. Five vegetation indices were also included to help distinguish between classes that exhibited similar spectral signatures. The optimization of DNN hyperparameters was conducted using three algorithms: Random Search, Hyperband, and Bayesian optimization. Overall, the Hyperband algorithm achieved the highest classification accuracy of 94.5%, which demonstrates the utility of optimized DL models for LCLU mapping when leveraging relevant ancillary metrics. Additionally, dropout regularization reduced DNN overfitting. The updated LCLU maps provided insights into land use inside the park, which was used to develop evidence-based conservation planning and management strategies in critical regions. This research exemplified the use of key ML applications to advance environmental monitoring via open-access cloud computing platforms and remotely sensed global datasets.

4.2. Review of DNN Algorithms for LCLU Classification

Previous studies have successfully employed similar approaches for LULC classification. Talukdar et al. (2021) reported an overall accuracy of 87% when distinguishing six LULC classes across India using a single-layer MLP (ANN) on Landsat 8 Operational Land Imager (OLI) imagery. A comparative analysis among six ML models in this study showed that the MLP model had the second-best performance after RF. Yuan et al. (2020) proposed combining unsupervised Kohonen's Self-Organizing Mapping neural network with a supervised DNN architecture consisting of three layers. They achieved an accuracy of 88.13% for classifying eight LULC classes using Landsat TM imagery in the Neuse River Basin region of North Carolina. Similarly, Camargo et al. (2019) attained an overall accuracy of approximately 73% when mapping nine classes in Brazil using polarimetric ALOS-2/PALSAR-2 L-band images. The comparative analysis of five classifiers byCamargo et al. (2019) helped categorize them into two groups. The first group, which obtained the best results, was comprised of RF, MLP and SVM algorithms, with statistically similar Kappa indices. The second group, consisting of NB and decision tree J48 classifiers, demonstrated less accurate performances and presented statistically similar Kappa indices.

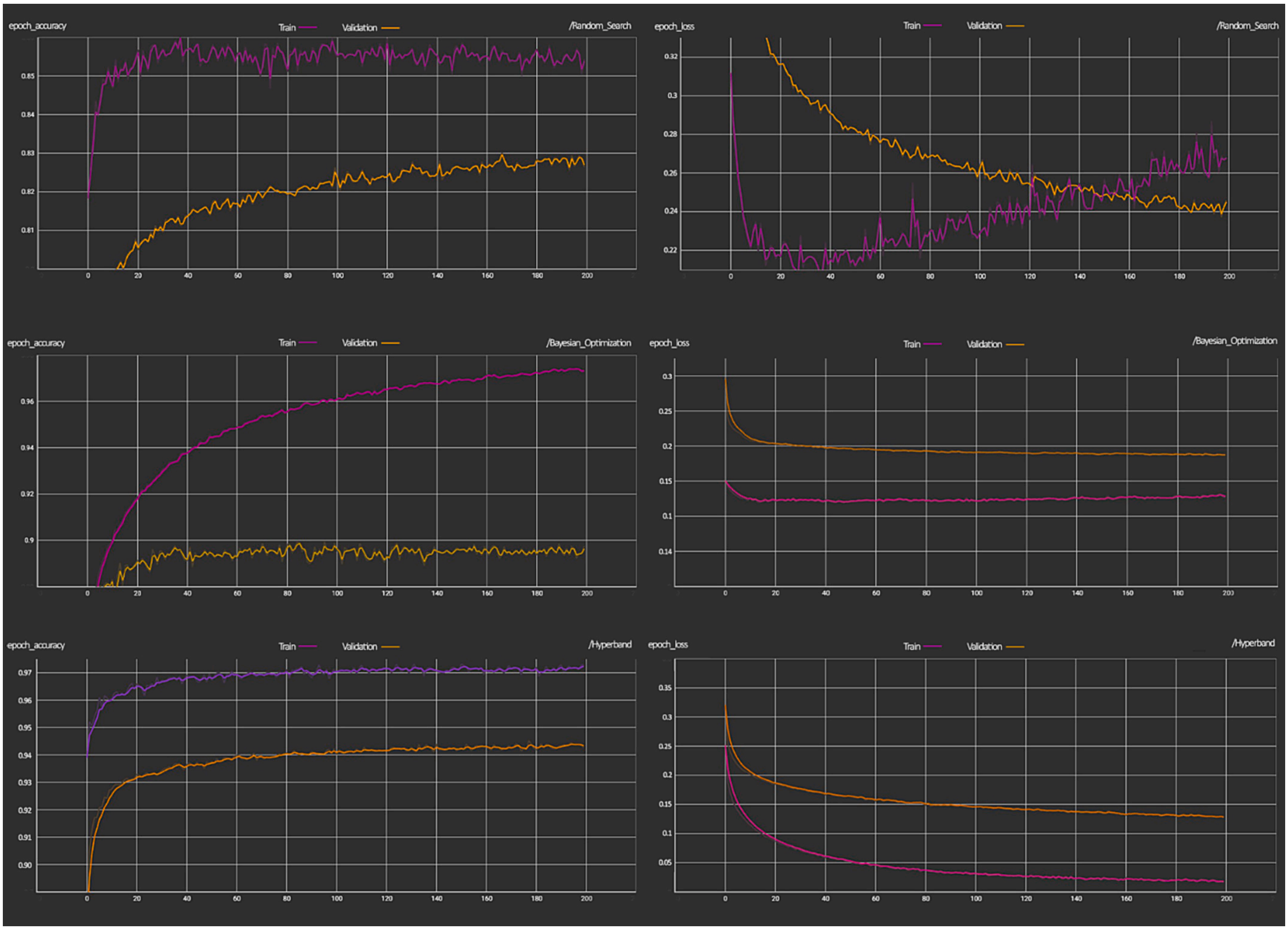

Fig. 7. Learning curve and loss function plots for the three DNN optimized architectures.

The literature demonstrates numerous successful applications of the DNN model for spatial patterns using LULC classification. In a comparative study, Thach et al. (2018) analyzed the spatial patterns of fire dangers in the tropical forests of the Thuan Chau district, Vietnam, using three ML algorithms: MLP, RF, and SVM. Global performance analysis showed that MLP achieved the highest predictive accuracy with an area under the curve (AUC) value of 0.894, followed by RF and SVM. Yousefi et al. (2021) employed MLP to identify optimal locations for *Juniperus excels* afforestation in Firuzkuh, Iran. MLP had the best performance with an AUC of 0.892. These studies highlight MLP's ability to extract meaningful spatial patterns from complex datasets and effectively model non-linear geographic phenomena. When combined with environmental data layers within a GIS framework, DNN has proven valuable for landscape investigations concerning habitat distribution, natural hazard susceptibility, and land management optimizations.

As an adaptive, non-linear statistical data modeling tool, DNN is well-suited for modeling the complex spatial interaction of biophysical and socio-economic factors that drive LULC changes over time. Numerous studies have demonstrated DNN's effectiveness for modeling historical LULC patterns and predicting future scenarios. Wang and Maduako (2018) integrated DNN with Markov chain modeling to achieve high predictive accuracy when simulating sustainable-urban planning in the Lagos Metropolitan Region and the Nigerian Federation. DNN was also combined with Markov processes to model informal wood harvesting in South Africa (Nel et al., 2017). Combining DNN and Markov has proven capable of distinguishing landscape dynamics and urban growth at finer scales in India (Bagaria et al., 2021; Devi and Shimrah, 2023; Sardar and Samadder, 2021), China (Xu et al., 2022; Yang et al., 2019), Saudi Arabia (Al-Shaar, 2023).

4.3. The Moroccan context

Recent studies have been conducted in Morocco using MLP for LCLU analysis (Boutahir et al., 2022; El Mustapha Azzirgue et al., 2022; Ouali, 2023). Saadani et al. (2020) conducted a study on the spatial–temporal monitoring of land use/cover change (LUCC) between 1999 and 2018 and urban growth modeling from 2010 to 2040 for El Jadida City, Morocco. Gadal et al. (2023) explored relationships between LULC and soil organic carbon stocks in Morocco by employing an MLP model to conduct geospatial modeling of soil organic carbon using environmental covariates derived from Landsat imagery and soil samples. The MLP approach achieved high positive autocorrelation between modeled carbon stocks and measured vegetation indices. Manaouch et al., (2023) evaluated the capabilities of six ML techniques (MLT), including MLP) for delineating optimal reforestation areas for *Quercus ilex* in upper Ziz, southeast Morocco. They found that Bagging and Naive Bayes models performed better in identifying areas suitable for reforestation located westward with low erodibility geology and adequate rainfall than DNN with AUCs of 0.98 and 0.97, respectively. On the other side, Benbriqa et al. (2021) evaluated ensemble learning techniques, including DNN used as a meta-learning model. They found it could outperform single learning classifiers when applied to the EuroSAT Moroccan LCLU dataset.

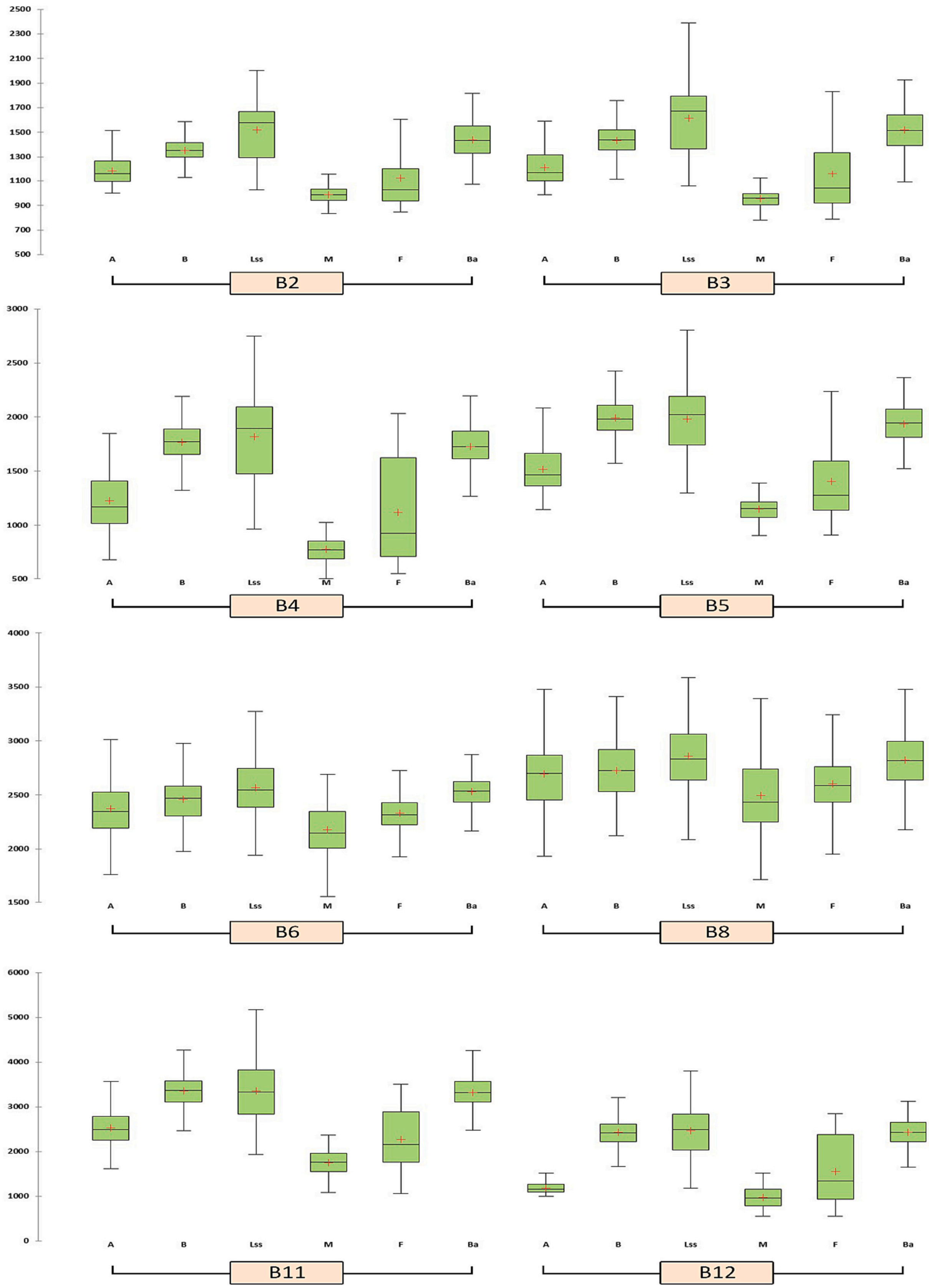

Fig. 8. Spectral reflectance variation of the LCLU classes (A: Agricultural lands; B: Bare lands; Lss: Limestone soil surfaces; M: Matorral; F: Forest; Ba: Build-up areas).

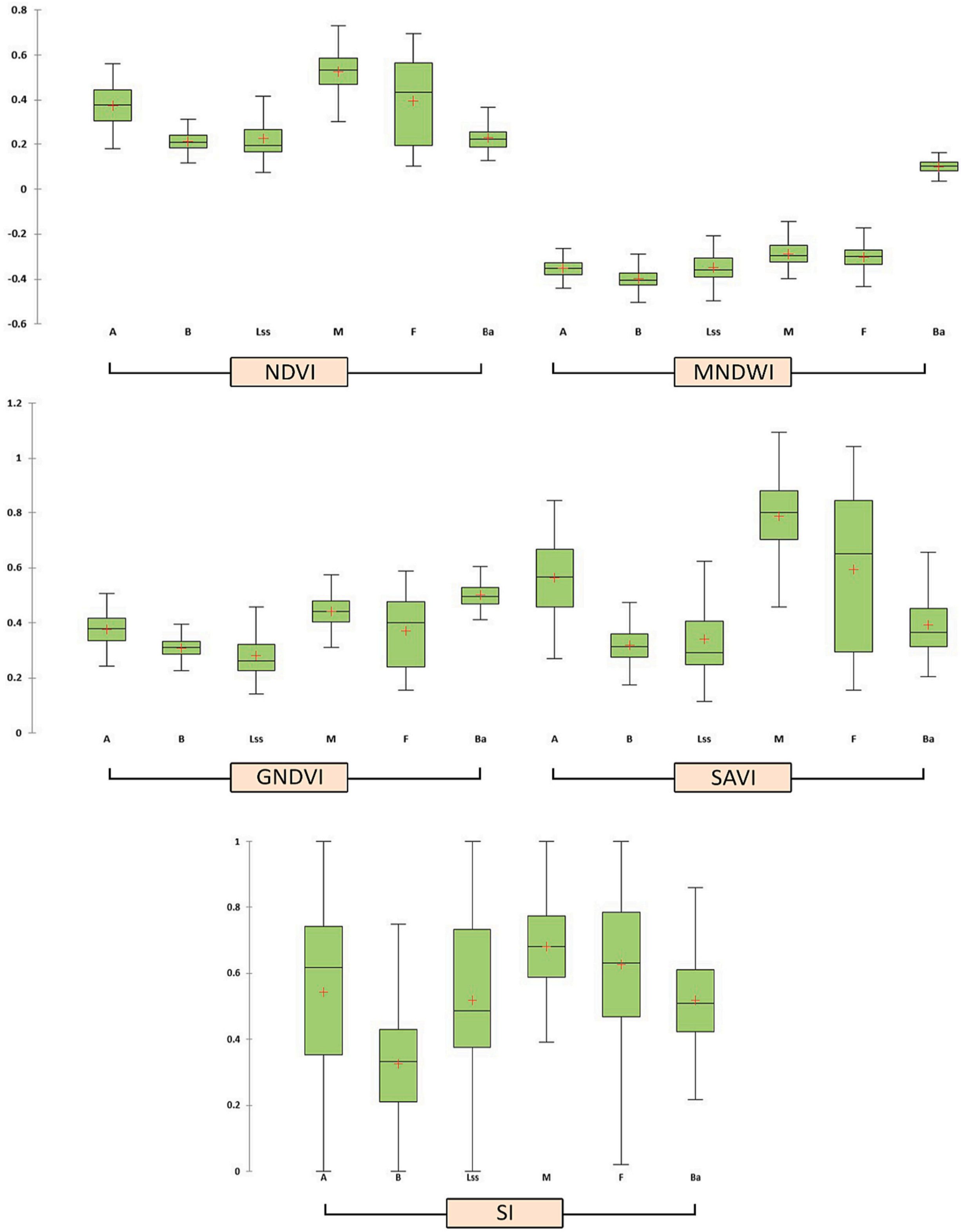

Fig. 8. (*continued*).

4.4. Overcoming LCLU classification challenges

LCLU classification is an important research area in RS; however, it faces various challenges that create significant data gaps. One challenge is achieving accurate classifications at local scales, as aggregated classes may hide important fine-grained information (Hoeser and Kuenzer, 2020). RS datasets also have spatial, spectral, and temporal resolution variability, making classification more difficult (Zhong et al., 2019). A further challenge is effectively mapping LCLU in areas experiencing rapid change, as traditional methods struggle with capturing and explaining these dynamics (Wang et al., 2022). Other issues include dealing with heterogeneous landscapes containing diverse land cover

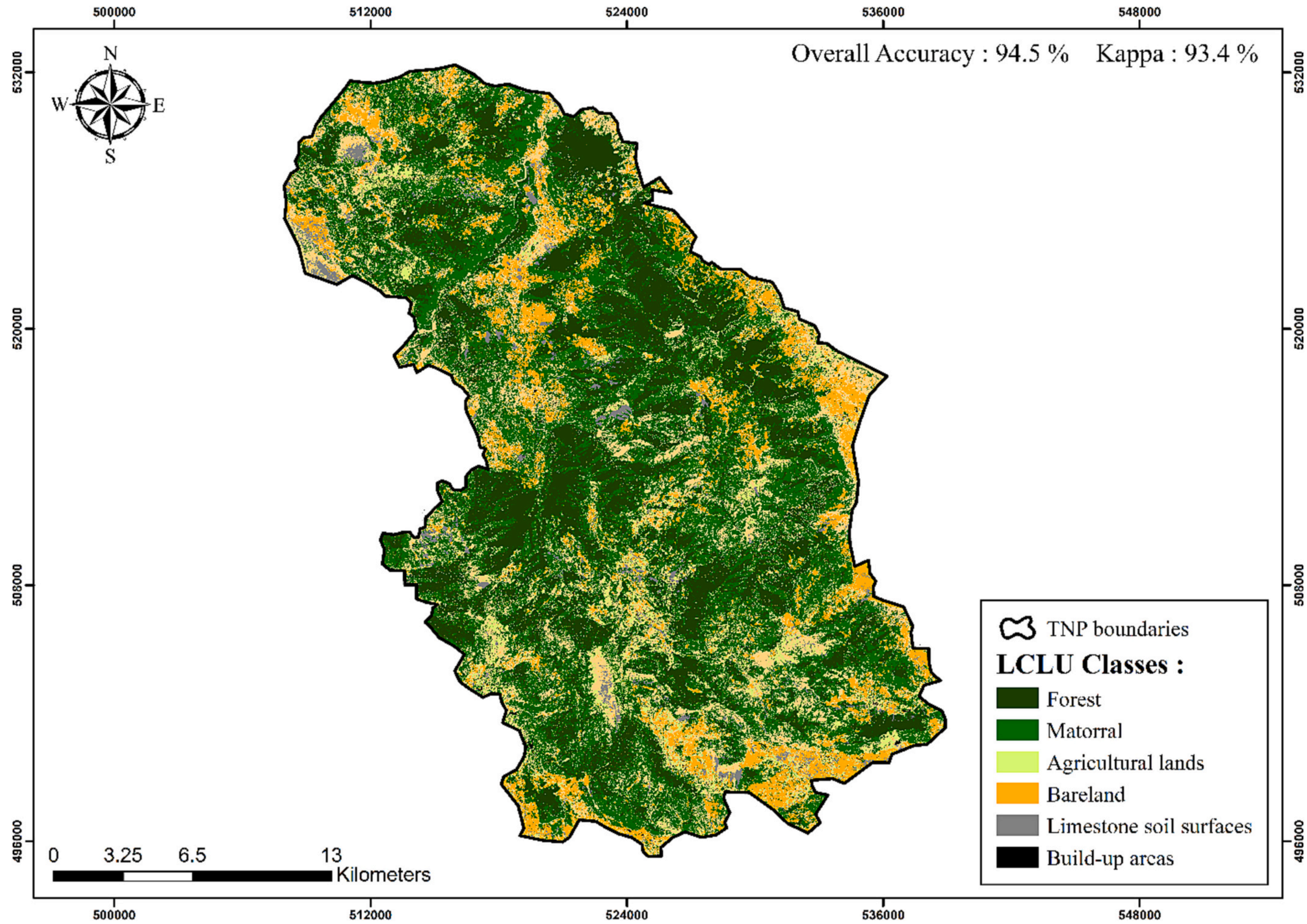

Fig. 9. LCLU map of the TNP for 2022.

types with similar spectral signatures, complex urban environments with mixed pixels, and ambiguous boundaries between categories (Sousa and Sousa, 2022; Pritchard et al., 2018). Additional gaps exist in classifying specific land cover types like non-forest tree plantations (Caughlin et al., 2021) and monitoring regrowth forests (Li et al., 2018). Increased data accessibility and finer spatial resolution will enable exploration of finer details, but will also require techniques that accommodate tiny or sparsely distributed objects while preserving contextual information (Neupane et al., 2021). Addressing these challenges and filling essential data gaps is needed to advance LCLU research using RS.

There are several limitations related to the methodology employed in this study. A primary concern is the selection and number of spectral indices used to classify land cover, as these features can strongly influence results. The five indices chosen—NDVI, GNDVI, SAVI, MNDWI, and NDWI—are commonly used based on prior research evaluating their performance for specific surfaces like vegetation or water bodies. However, their transferability across different contexts with diverse land cover characteristics cannot be guaranteed. In our context, it is very difficult to detect the built-up areas because they are tightly grouped and built of stone, adobe, or rammed earth on terraced hills. This led to similarities in reflectance between built-up areas and neighboring limestone surfaces, causing uncertainties in our model. In addition, NDVI and GNDVI are most sensitive to vegetation greenness but may not capture non-green attributes equally well. Similarly, SAVI and MNDWI aim to correct limitations of other indices for soil and water, respectively, but their effectiveness varies across sites. Additionally, other less explored indices could potentially improve the representation of specific land cover types not optimally separated by the chosen indices. Lastly, the number of indices used represents a tradeoff, as more features increase complexity and demand for training data.

Another critical limitation relates to the use of DNN for LCLU classification. Although DNN, a variant of ANNs, has emerged as a powerful tool for pattern recognition (Gardner and Dorling, 1998) and pixel-wise classification (Taravat et al., 2015), it has certain limitations. The architecture, including the number of layers and neurons in each layer, of a DNN model highly impacts performance but is often determined heuristically without guaranteeing optimal performance. Furthermore, DNN is sensitive to the number of hidden layers and neurons, which can lead to overfitting or underfitting of the data. DNN's sensitivity is also linked to various factors such as sample size, degree of non-linearity, noise, and missing values in the underlying problem (Walde et al., 2004). Lastly, DNN models are prone to overfitting (Lawrence and Giles, 2000) when dealing with high-dimensional data, which was mitigated using dropout regularization. However, this method might not fully prevent overfitting, and it can sometimes lead to underfitting if overused.

Our study employed three popular hyperparameter optimization approaches: Random Search, Hyperband, and Bayesian Optimization. Each method has its own strengths and weaknesses. For instance, although Hyperband improved our classification accuracy, it involved making several assumptions about resource allocation that may only hold in some scenarios. Moreover, although Bayesian Optimization provided a systematic approach towards hyperparameters tuning, it can be computationally expensive and may not always converge to the global optimum (Snoek et al., 2012). Several studies that combine these

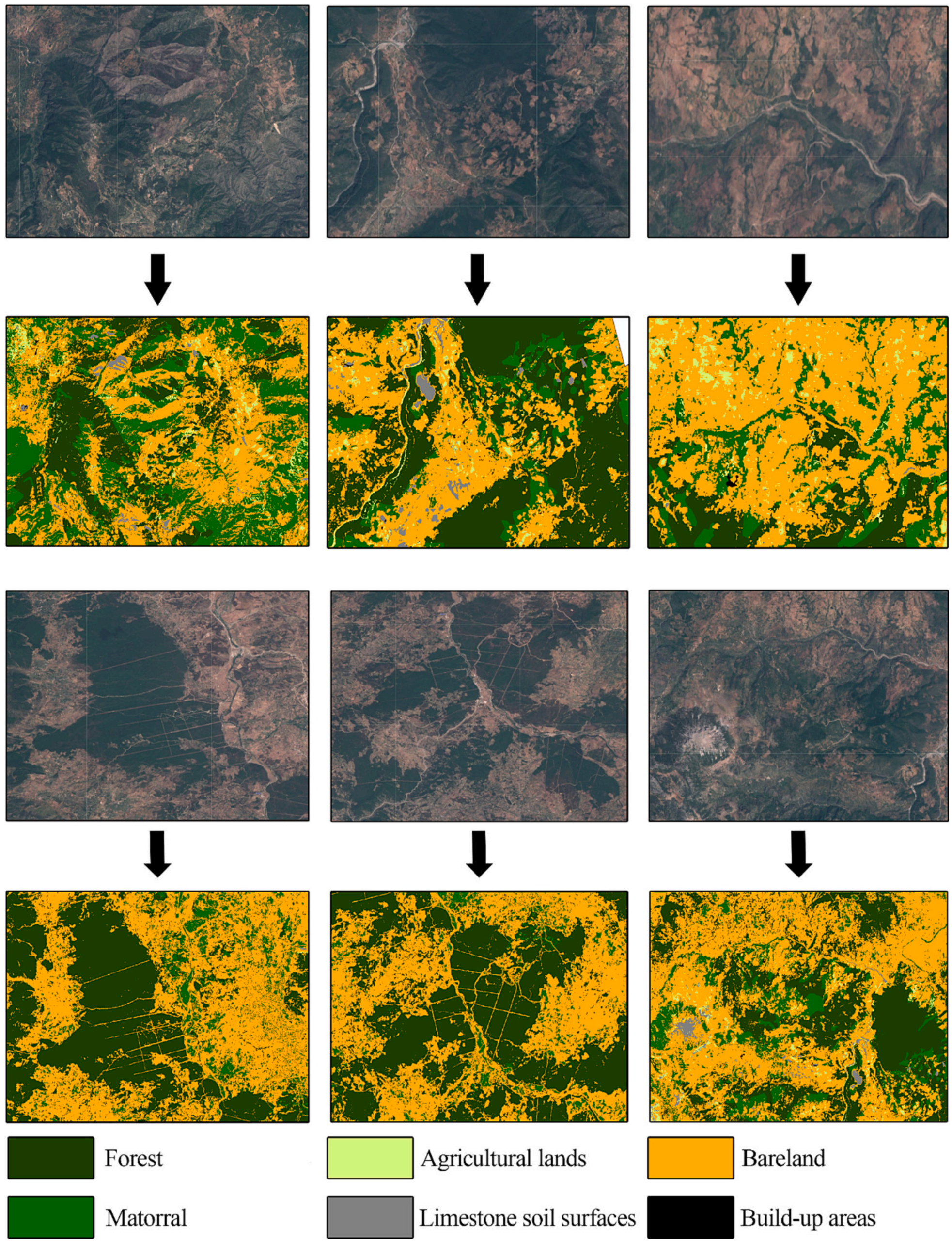

Fig. 10. Detailed illustration of the classification results for typical areas of the TNP.

Table 3
Areas of each LCLU class in the TNP for the year 2022.

	Surface (ha)
Agricultural lands	2052.359695
Bareland	28,010.96819
Build-up areas	723.6793209
Limestone soil surfaces	1048.097338
Forest	14,960.33338
Matorral	13,873.56207

methods to leverage their complementary strengths also emerged. Wang et al. (2018a, 2018b) combined Bayesian optimization and Hyperband approaches, which showed that the combination approach outperformed other hyperparameter optimization methods, including the Hyperband algorithm. Other work focused on further developing the Hyperband algorithm by incorporating the Bayesian optimization method that captured the density of favorable configurations in the input space (Falkner et al., 2017). This new method combined the anytime property of Hyperband with the guided approach of Bayesian optimization to identify the best configuration.

Our study faced similar challenges gathering ground truthing data as previous studies. The accuracy of the LCLU classification is heavily dependent on the quality and quantity of ground truth data. As these data were annotated through visual interpretation and validated through aerial and satellite imagery field surveys, there is inherent uncertainty compared to field survey data. The degree of geolocation errors, inconsistencies in class definitions, and ambiguities at class boundaries introduced during the labeling process are difficult to quantify. Using limited reference samples also means some classes may not be captured with full variability. These challenges are common in RS studies where extensive field surveys are infeasible. Other studies have also noted issues arising from reference data limitations. Brown et al. (2022) found significant discrepancies between expert and non-expert

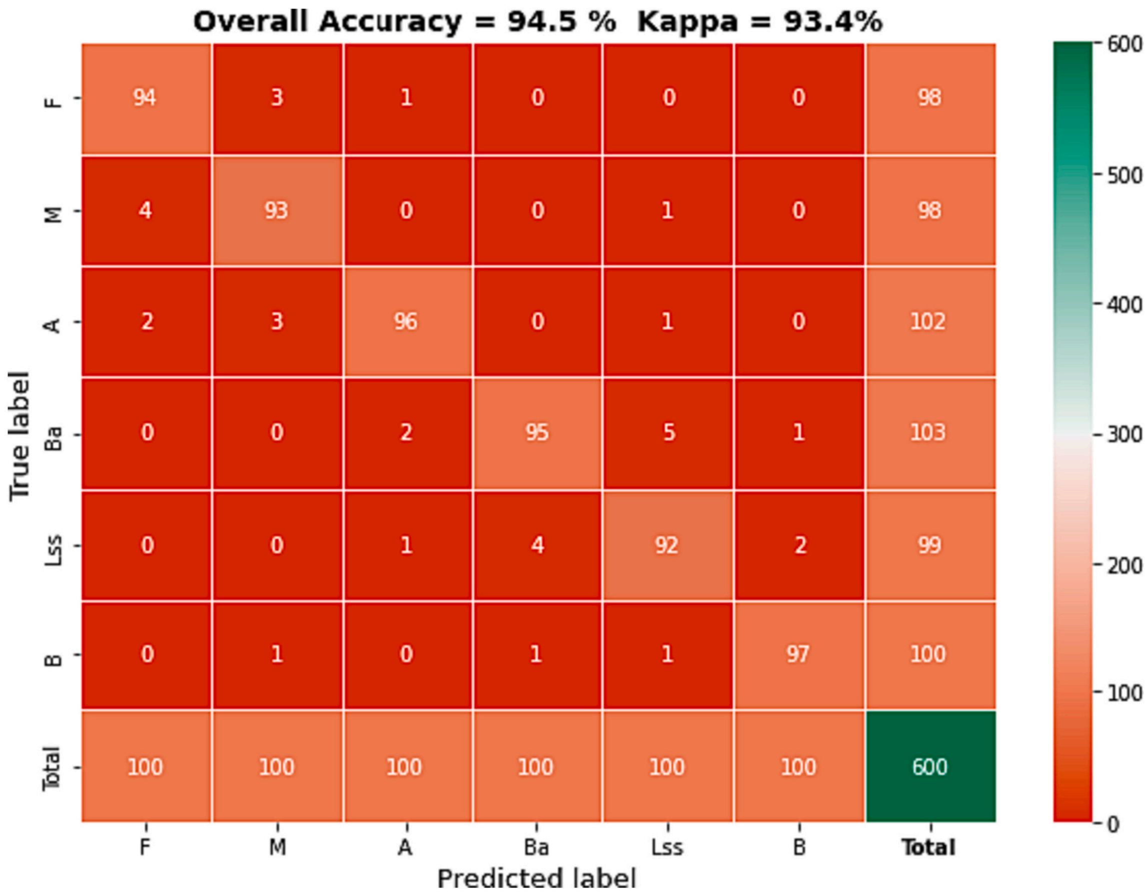

Fig. 11. Confusion matrix Heatmap of the test dataset (A: Agricultural lands; B: Bare lands; Lss: Limestone soil surfaces; M: Matorral; F: Forest; Ba: Build-up areas).

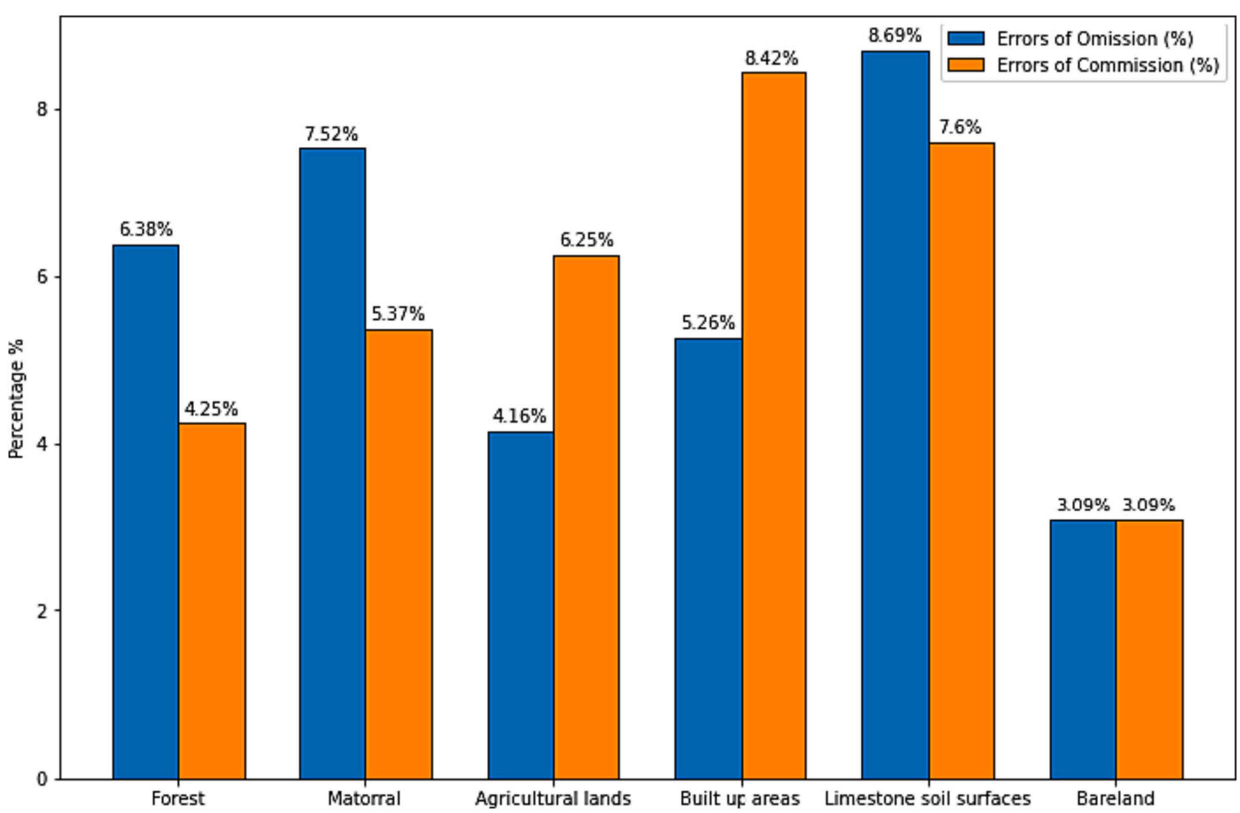

Fig. 12. Errors of omission and commission of our LCLU classes.

image annotations. A pixel-wide comparison of labeling precisions revealed producer's accuracies as low as 22% for grassland and 31% for bare ground surfaces in non-expert classifications relative to expert delineations. It was also found that OpenStreetMap labels contain errors requiring manual corrections (Johnson and Iizuka, 2016). Elmes et al., (2020) reviewed issues with reference datasets like spectral variability within classes, class representation biases, ambiguous category boundaries, and labeling inconsistencies over large areas. Developing techniques robust to such reference data issues remains an important focus, as uncertainties can propagate through the trained DNN and degrade performance. Larger, more consistently annotated datasets would help address these challenges, but generating such data at a meaningful scale remains resource-intensive. In summary, the reference truth provides essential supervision but contains vagaries that impact modeling, highlighting a need for techniques that can overcome these uncertainties.

4.5. Outlook and further research avenues

This study demonstrated the promise of using a DNN model for LCLU mapping; however, more comprehensive validation is needed to fully realize the potential of DL approaches for LCLU mapping. Future research could focus on conducting a complete stand distribution assessment by investigating the effectiveness of other advanced DL architectures for this application. This could involve evaluating convolutional or recurrent neural networks on diverse datasets to identify models best suited to different mapping contexts or scales. Incorporating ancillary data and propagating uncertainty from reference points may also enhance classification reliability. Examining error sources through detailed confusion matrices could provide additional insight to refine categories, features, or algorithms. Combining multi-source RS data holds promise to augment spectral information as well. Overall, continued testing of model robustness, refinement through uncertainty analysis, and integration of environmental context would help optimize land cover mapping with DL techniques. Addressing these opportunities would move the field closer to establishing transparent and reliable techniques supporting natural resource management needs.

5. Conclusion

Accurate and timely LCLU assessment is essential for effective and timely natural resources management. This study combined AI algorithms in a cloud-based environment to identify the type, extent, and distribution of different LCLU types within the TNP area for 2022. The aim was to test the potential of the MPL model by comparing hyper-parameter optimization methods to achieve the highest overall accuracy. Given its importance, the model's optimization architecture was carefully considered to improve the performance and interpretability of the DNN model.

Classification results showed that using five spectral indices enabled better distinguishment between similar surface classes. With an overall accuracy of 94.5% and a kappa coefficient of 93.4%, the classification accuracy of our DNN model was excellent. Comparison between optimization algorithms showed that Hyperband was the fastest method and achieved the highest overall accuracy. In contrast, Bayesian optimization achieved slightly lower accuracy but was still significantly higher than Random Search. When high accuracy is a priority, and computational resources are not limited, Hyperband proved to be the best optimization approach. Finally, implementing more sophisticated classification algorithms can improve the classification, monitoring, and management of forest resources, especially on small spatial scales, and assist decision-makers by providing actionable insights based on accurate data.

Declaration of Competing Interest

None.

Data availability

Data will be made available on request.

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